Acrome Education Kit Code Helper and Control

What is SMD Education Kit?

The Education Kit is designed as an educational-focused robotic learning platform. This kit is offered as a training tool for students, academics, and engineers to learn about robotics and automation systems.

With the Education Kit developed by Acrome, it is possible to understand the fundamentals of engineering while also providing an opportunity to test and practice advanced control algorithms. It is seen as an ideal solution, especially for learning the working principles of motors, sensors, and other components used in robotic systems.

What is the application and what does it do?

This application integrates artificial intelligence to generate code, with the chatbot serving as an interface for asking questions and writing code. Both the chatbot and the Education Kit Control sections generate code, but they function in different ways. The chatbot interface allows users to write code directly on the Tkinter page, where the generated code remains within the interface, offering immediate help and suggestions.

On the other hand, the Education Kit Control section not only generates code but also saves it as a .py file and executes it. When the "Command" button is pressed or when "Change the Camera" is selected in the Education Control section, the camera is activated. This allows users to observe and interact with the system in real-time, providing a hands-on experience with the Education Kit and its components.

Thus, the application provides two distinct functionalities: one that enables code generation and immediate assistance through the chatbot interface, and another that saves and runs the generated code via the Education Kit Control section. This combination allows users to experiment with robotics and automation systems both by generating code in real-time and interacting with the physical components of the Education Kit.

Steps of Application

1. Enter API Keys

First, when you enter the application, you see this screen.

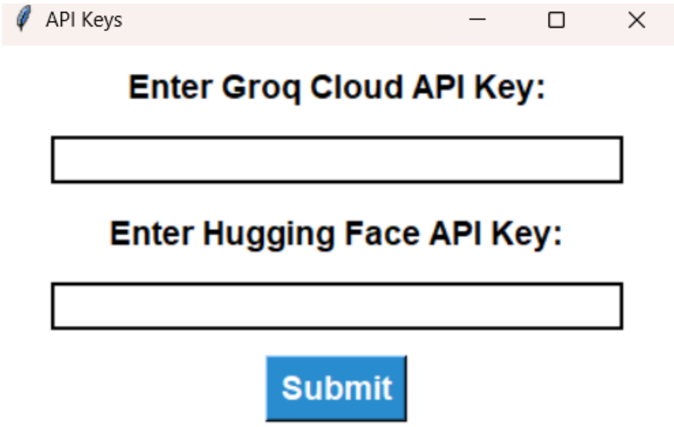

Figure 1: API Screen

You need to fill in the API fields. For this, you need to get an api key from Groq Cloud and Hugging Face.

For Groq Cloud:

Visit the following site: Groq API Documentation**.**

Click on "API Keys." If you are not logged in, you will see a screen like this: Here, you can log in using GitHub, Google, or by creating a regular account.

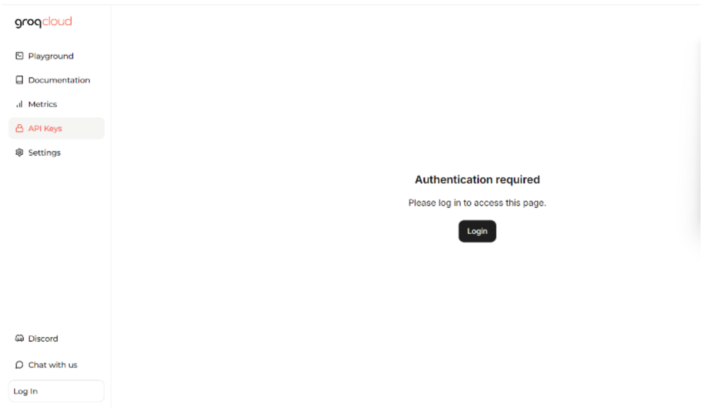

Figure 2: Groq Cloud

If you are already logged in, the interface will look like this:

Click on "Create API Key," give your API key a name, and generate it.

Important:

Make sure to save your API key, as it is typically shown only once and cannot be retrieved again.

If you accidentally delete the file containing the API key, you may encounter issues. Because .exe reads API Keys in .txt file.

The solution in such cases is to generate a new API key by following the same steps.

For Hugging Face API:

Go to the following link: Hugging Face.

If you already have an account, log in. If you're new, you'll need to register first.

After logging in, click on your profile picture in the top-right corner.

From the dropdown menu, select Access Tokens.

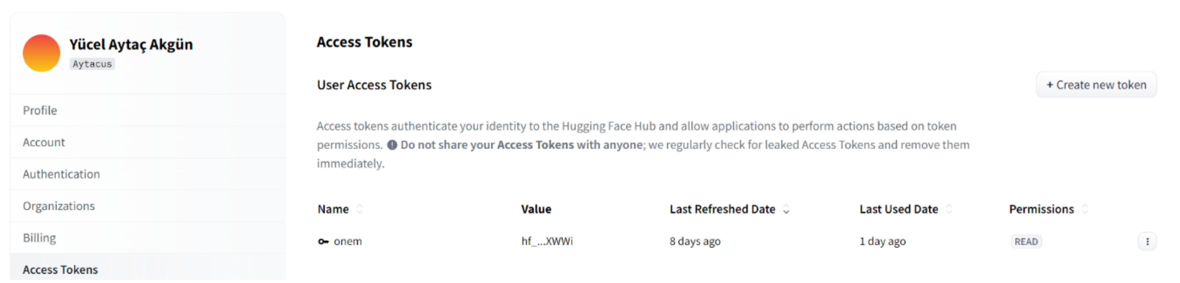

Figure3: Hugging Face Access Tokens

Click on Create New Token to generate an API key.

You will be prompted to select one of three token types: Read, Write, or Fine-grained.

Choose the Read option, enter a name for your token, and complete the API creation process.

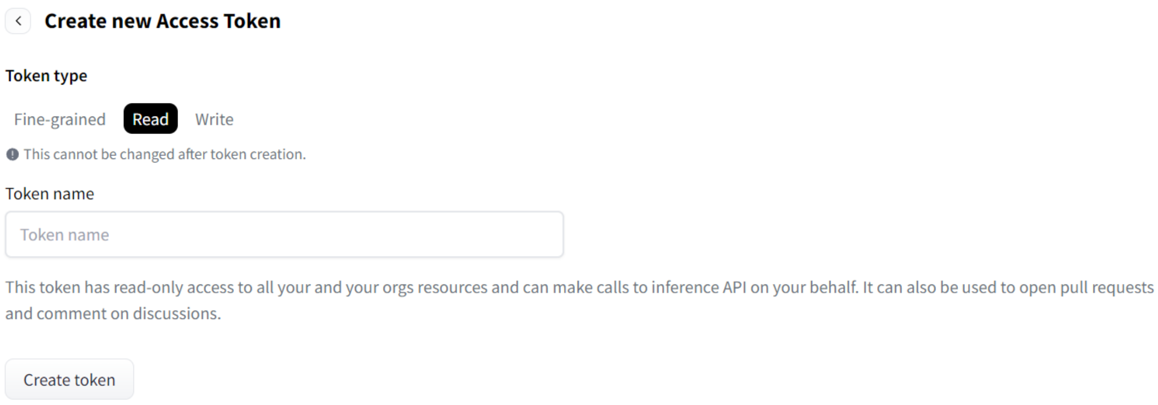

Figure4: Create new Access Token screen

Once the token is created, make sure to save it securely, as it will be needed for integration with the application.

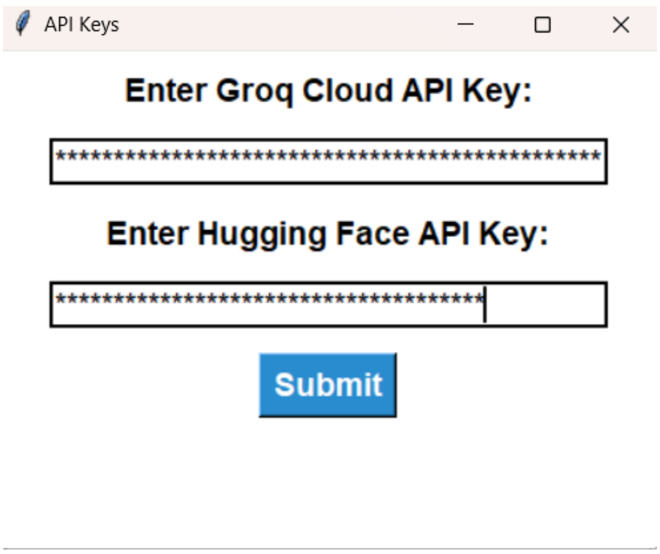

Figure5: Final Api Screen

When Click Submit, you will pass main application. If API keys are disable or wrong, you cannot use main app because of error. Also these keys save .txt file so if you delete txt file, you can see API Keys screen again. But you have txt file, you can run main app directly.

2. Chatbot Side:

If you use chatbot, you will see this screen.

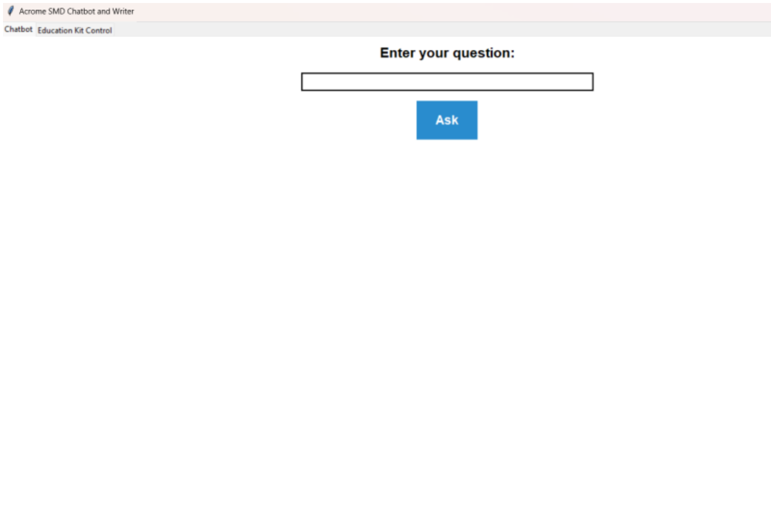

Figure 6: Chatbot Screen

Purpose of Chatbot Section

The main purpose of this section is to allow users to input prompts and press the **Ask** button to receive detailed information or answers, making it an ideal tool for learning and exploring topics. The chatbot is designed to facilitate an interactive and engaging experience, where users can ask questions, seek clarification on concepts, or request code snippets to address specific tasks.

By leveraging artificial intelligence, this feature provides accurate and contextually relevant responses to user queries, making it particularly useful for those looking to deepen their understanding of robotics, automation systems, or programming. The simple and intuitive interface ensures that users can focus on their learning goals without needing prior technical expertise.

In essence, this section serves as an educational resource that combines convenience, accuracy, and accessibility to help users expand their knowledge effectively.

3. Education Kit Control:

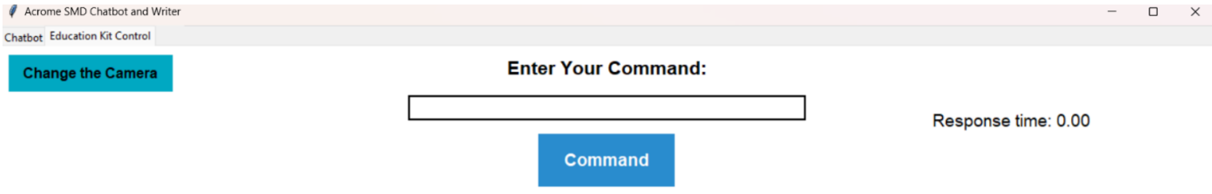

Figure 7: Education Kit Control Screen

Purpose of Education Kit Control Section

The "Change the Camera" button allows users to switch between internal and external cameras. If no internal camera is available or no external camera is connected, an error message will appear stating, "Camera is not available or not connected." This message is purely informational and does not cause the application to crash; it simply means the camera cannot be used until a valid device is connected.

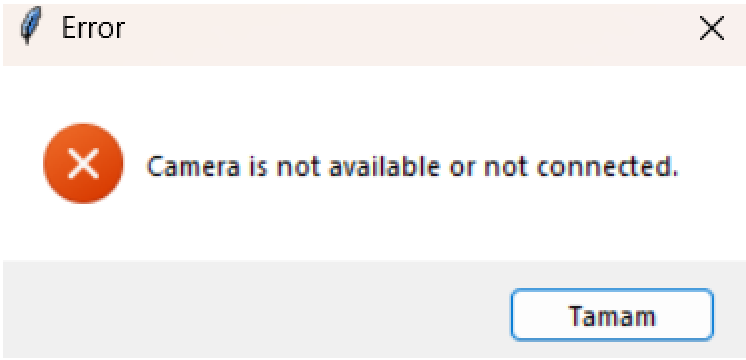

Figure 8: Error for camera

When you click "Change the Camera," the application toggles between available cameras, and the selected camera will stay active for 10 seconds before automatically turning off. This short duration helps the user identify which camera is currently in use.

Additionally, pressing the "Command" button performs two actions: it activates a camera for 60 seconds and writes Python code to a file, then executes that file. This feature not only provides a dynamic interaction with the system but also ensures efficient resource management by preventing the camera from being continuously active.

Additionally, if the Python code executed through the "Command" button fails to run, the application provides an error message indicating why the execution failed. This feature helps users quickly identify and resolve issues, ensuring a smoother and more efficient workflow.

Like this:

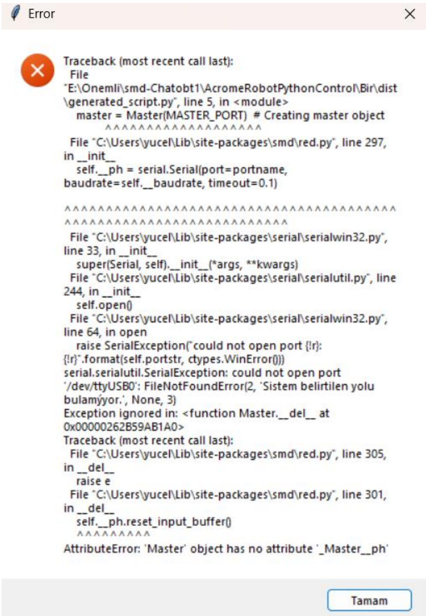

Figure 9: Error message for Python

In summary, these features provide flexibility in camera selection and usage while ensuring the system remains stable and user-friendly, even when no camera is connected.

IMPORTANCE NOTE:

- For the application to work correctly, it must have API Keys.
- You should open a “data” (folder name) folder in the location where the exe is located and paste it there as documentation (txt).

Have a Fun 😊